

40m 'Pebble' HF Radio Construction and Operating Manual

Version 1.0



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Version 1.0

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Initial Release (still in draft form)

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This construction and Operating Manual has been written specifically to support the Group Buy/Build of the 40m Pebble HF radio by the Manly Warringah Radio Society (MWRS) in Sydney Australia. <https://www.mwrs.org.au/>
The manual draws on the writer's experience building and testing the 40m variant v2.7, as well as pointing to and drawing from other Pebble resources available on the Internet.

The 40m Pebble kit is not sold commercially by the MWRS nor the author of this document. Those wishing to source a Pebble should look initially to those being sold by HamRadioDuo (<https://hamradioduo.com/>) who at the time of writing were primarily supplying radio's to the USA/Canada Market.

Open Source

This project is an open-source project and community supported. Please join the Discord to chat with other Pebble HF owners to share in helping each other.

Hardware open source: github.com/WB2CBA/PEBBLE-20m-HF-TRANSCEIVER

Firmware open source: github.com/mspiceland/usdx-pebblehf

A huge vote of thanks and appreciation for those who have put so much work into developing the Pebble Project for the Amateur Radio Community. Your efforts are greatly appreciated and valued. THANKYOU ☺

Hardware Design: **Barb WB2CBA**

Firmware: **Based on original uSDX firmware by Guido PE1NNZ ; adapted by Mike N4FFF**

Introduction

The Pebble HF Radio is a single band, QRP Transceiver for Ham Radio. Designed as an ultra-inexpensive radio to encourage more people to have fun with HF and outdoor ham radio operations. The radio has the following characteristics:

- Easily built and inexpensive single band transceiver
- CW, SSB Voice (USB and LSB), and digital modes (using an adapter)
- Built in CW key
- Built in microphone for voice
- Based on the uSDX open-source project, biased for cost and improvements for CW

Hardware Design: **Barb WB2CBA**

Firmware: **Based on original uSDX firmware by Guido PE1NNZ ; adapted by Mike N4FFF**

At the time of writing these notes, the 20m Pebble was the only variant officially released, but I was fortunate to receive construction files for a 40m variant of the Pebble from the designer Barb WB2CBA) so that we could develop and promote the use of the Pebble through the MWRS Portable Special Interest Group, primarily during their Friday Morning Park activities. The 40m Pebble is ideally suited for this application.



The 'Pebble' operating portable – power supplied by a 3x18650 battery pack (notionally 12.6v) and shown here using an external amplified speaker mic (although on the speaker was being used) and connected to a small ATU which was monitoring the power output and SWR of an EFHW antenna during the initial field tests. The ATU is not needed if you are connected to a resonant antenna.

The radio produces a solid 4.5w – 5w output using a 12v battery with a very efficient Class E power amplifier, or 1w output if powering from a 5v USB power source – ideal for WSPR or Digital operation.



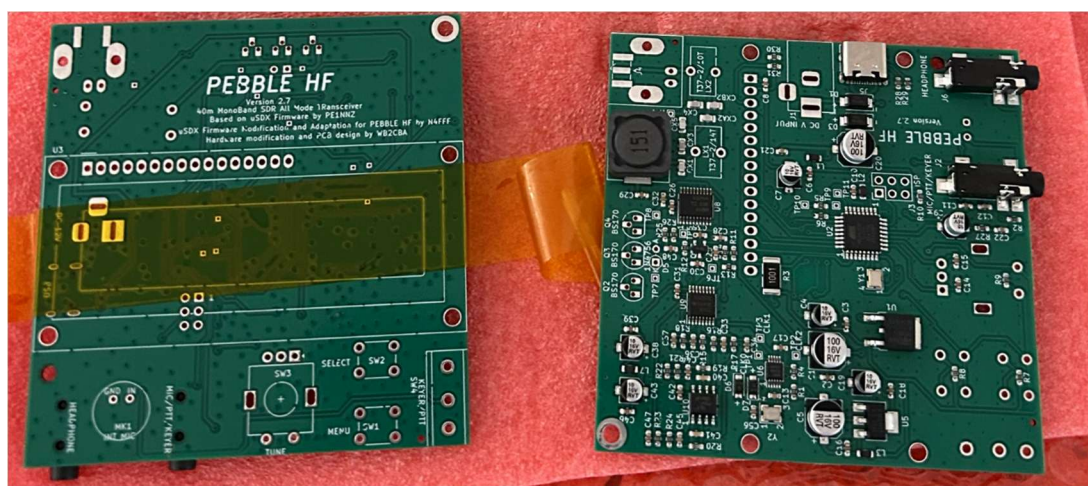
The Pebble makes an easily transported HF radio and when connected to a simple wire antenna (eg: dipole or EFHW) provides a fun, capable and enjoyable field radio experience, with very little financial investment. Good quality audio for SSB is achieved using the onboard electret microphone (or external mic if required) together with a speaker or headphone via the audio socket.

CW operation is supported with an inbuilt CW decoder, and use of the inbuilt CW key of by plugging in a straight or iambic paddle.

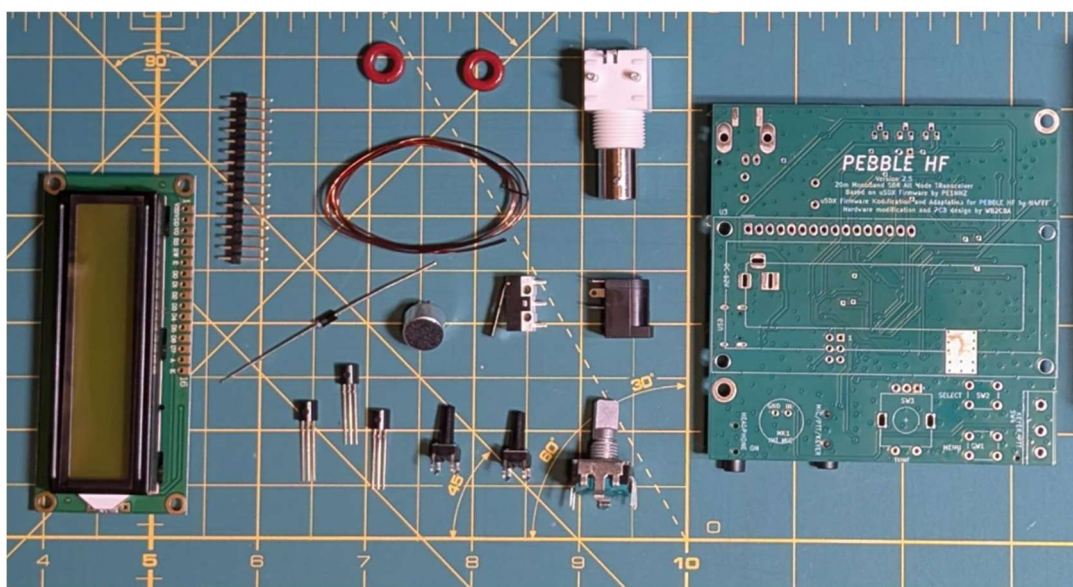
Digital modes are supported via the audio out and audio in sockets, and VOX control for triggering Digital Tx,

The Pebble is largely prebuilt with all the SMD components preassembled. There are two inductors to be wound onto a T37-2 toroid, and together with a small number of through hole devices to be soldered onto the PCB, most constructors will be able to build the radio in under an hour. Software is loaded onto the PCB using the Arduino IDE with an Arduino Uno (or similar) acting as an ISP, after burning the bootloader onto the chip using the same method.

The photo below shows the main PCB v2.7 for 40m (front and back) as received from the factory with the preassembled components



The photo below shows the additional components required to be added to the main board to complete the build. The photo is taken from the pebble construction manual for the 20m version, but from a constructor's point of view the only real change to the number of turns to be wound on the T37-2 toroids.



About the Radio

This description was found in the mspiceland GitHub and provides a great overview of the Pebble. The original details can be found here: <https://github.com/mspiceland/usdx-pebblehf> I encourage you to visit the site for future updates and information.

“Amateur radio is a fantastic hobby for learning, connection, and adventure — and small, portable radios are perfect for getting on HF, connecting with people around the world, and outdoor radio adventures like POTA and SOTA.

Pebble HF was purpose-designed to be ultra-affordable and easy to build. It is a single-band 20-meter QRP transceiver that supports CW (morse code), SSB voice (USB/LSB), and digital modes such as FT8, JS8, and FT4. The hardware was designed from the ground up by Barb WB2CBA with cost and kit-friendliness in mind — nearly all parts are pre-soldered surface mount, leaving only 14 easy through-hole parts for the builder. Most people complete the build in under an hour.

Despite its simplicity, Pebble HF is a capable software-defined radio. Both the transmitter and receiver run on a single ATMEGA328P microcontroller. The transmitter uses an efficient Class-E power amplifier driven entirely by digital signal processing, and the receiver implements DSP filtering, AGC, noise reduction, and more — all in software. This keeps the parts count low, the design simple, and the radio highly power-efficient.

- **~5W output** on 7–14V battery power, **~1W on USB-C (5V)** — get on the air even without a dedicated battery
- **Weighs less than 4.7 oz (132g)** — perfect for portable operation
- **Built-in CW key and microphone** — nothing extra needed to start making contacts
- **Open source hardware and firmware** — hack it, improve it, share it

Our mission is simple: get more people into HF ham radio and outdoor radio!

List of Features

- **Simple, fun and versatile QRP SSB HF transceiver** with embedded **DSP and SDR functions**
- **20-meter single-band** operation (14.000 - 14.350 MHz) (**MWRS Version is 40m (7.000 – 7.300 Mhz)**)
- **EER Class-E** driven SSB transmit-stage
- Approximately **5W PEP SSB output** from 13.8V supply
- **Mode support: USB, LSB, CW**
- **DSP filters: Full, 3000, 2400, 1800, 500, 200, 100, 50 Hz passband**
- **DSP features: Automatic Gain Control (AGC), Noise-reduction (NR), Voice-triggered Xmit (VOX), RX Attenuators (ATT), TX noise gate, TX drive control, Volume control, dBm/S-meter**
- SSB opposite side-band/carrier suppression **Transmit: better than -45dBc, IMD3 (two-tone) -33dBc, Receive: better than -50dBc**
- **Open source** firmware, built with Arduino IDE; allows experimentation, new features can be added, contributions can be shared via Github, software-complexity: 2000 lines of code
- Software-based **VOX** that can be used as **fast Full Break-In** (QSK and semi-QSK operation) or assist in RX/TX switching for operating digital modes (no CAT or PTT interface required), external PTT output/PA control with **TX-delay**
- **Lightweight and low-cost transceiver design:** because of the EER-transmitter class-E stage it is **highly power-efficient** (no bulky heatsinks required), and has a **simple design** (no complex balanced linear power amplifier required)

- **Fully digital and software-based SSB transmit-stage:** samples microphone-input and reconstruct a SSB-signal by controlling the phase of the SI5351 PLL (through tiny frequency changes over 800kbits/s I2C) and the amplitude of the PA (through PWM of the PA key-shaping circuit)
- **Fully digital and software-based SDR receiver-stages (optionally):** samples I/Q (complex) signal from Quadrature Sampling Detector digital mixer, and performs a 90-degree phase-shift mathematically in software (Hilbert-transform) and cancels out one side-band by adding them
- **CW decoder, Straight/Iambic-A/B keyer**
- **VFO A/B + RIT**, and corresponding relay band-filter switching via I2C
- Probably the most **cost effective** and **easy** to build standalone SDR/SSB transceiver that you can find. Very much **simplified** circuit and **versatile** in use.

Hardware

The Pebble HF hardware is designed by [Barbaros Asuroglu, WB2CBA](#). For hardware schematics, bill of materials, and build information, see [Barb's hardware repository on GitHub](#). For general project information, visit pebblehf.com.

This project is based on the original uSDX design. For other uSDX hardware variants, see the [uSDX Forum](#).

Operation

Pebble HF has three controls: a **tuning knob** (rotary encoder), a **MENU button** (left), and a **SELECT button** (right).

- **Tune:** Turn the encoder to change frequency (locked to 14.000–14.350 MHz) or (7.000 – 7.300 Mhz).
- **Volume:** Press and hold the encoder while turning. Turning below 0 powers off the radio.
- **Step size:** Short press encoder to increase, long press to decrease.
- **Filter:** Double-click SELECT (right) to cycle through DSP filter bandwidths.
- **VFO/RIT:** Long press SELECT (right) to toggle RIT, then again to swap VFO A/B.
- **Menu:** Short press MENU (left) to enter. Turn encoder to browse, press MENU to edit a value, press SELECT to save/exit.
- **Factory reset:** Hold encoder button during power-up.

Pebble HF has a built-in straight key and microphone. An external key or mic can be connected to the key/mic jack. Plugging something into the jack disables the built-in mic. **Always have a paddle plugged in when using Iambic A/B keyer mode** — otherwise the onboard mic may falsely trigger the keyer.”

Kit Assembly

Bill of Materials (BOM)

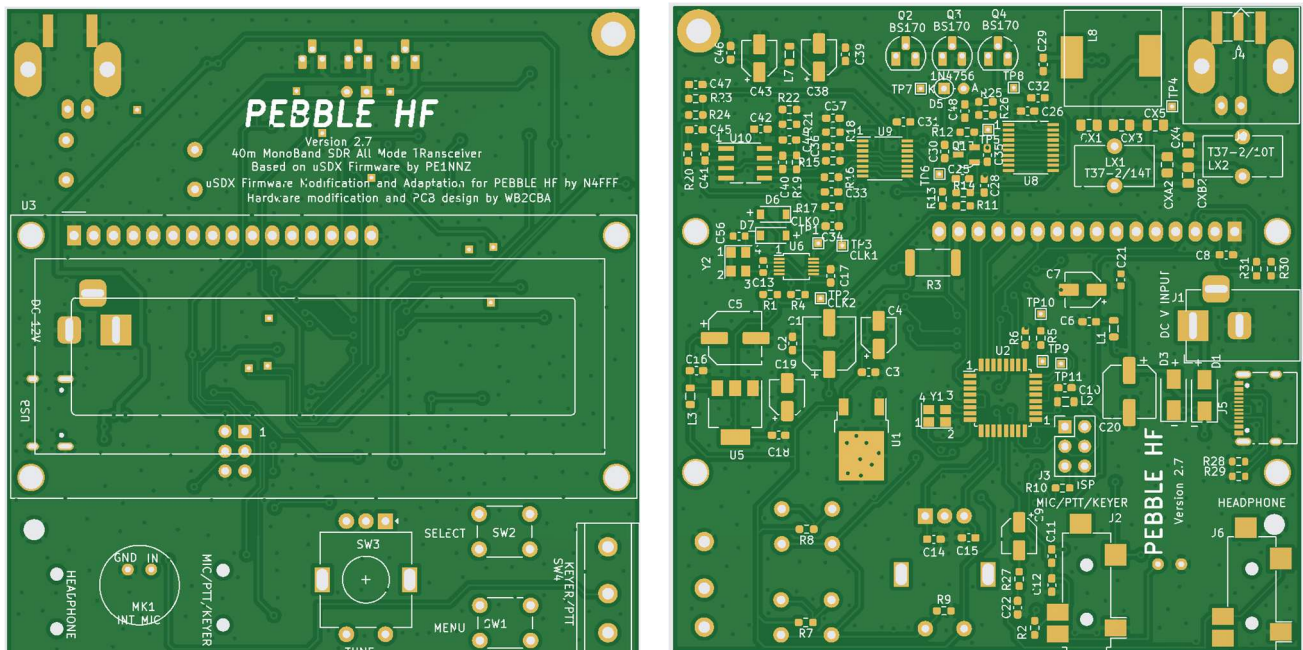
The following BOM is that provided with the MWRS 40m Pebble Group Buy Kit

Item	Desc	Qty
D1	1N4756	1
Male Pin Header 2.54mm	1 x 12, 2 x 3 Pin Header 2.5mm strips	18 pins
J2	DC_Socket_DC-005 and Plug	1
Display	16 x 2 LCD Display	1
Power Transistors	BS170 use known original sourced transistors	3
Rotary Encoder	PEC11R-4015F-S0024	1
SW1 , SW2 Z(6x6x13) longer shank if reqd	6mm vertical push button	2
SW3	SW_Slide_DPDT (On/Off) SK22D02	1
Key - Micro Switch	AC 125V 1A SPDT 1NO 1NC Momentary Long Hinge	1
BNC	BNC Antenna Socket	1
Microphone	Microphone 9x7mm with Pins	1
Toroids	T37-2	2
Wire	Insulated Copper 0.5mm dia (to be cut to 170mm and 230mm for inductors)	0.5m
Knob to suite Rotary Encoder Shaft	Eg: A01 20mm x 12mm x 6mm hole diameter	1
PCB - 40m Pebble HF Radio PCB	Pre Assembled Pebble 40m PCB	1

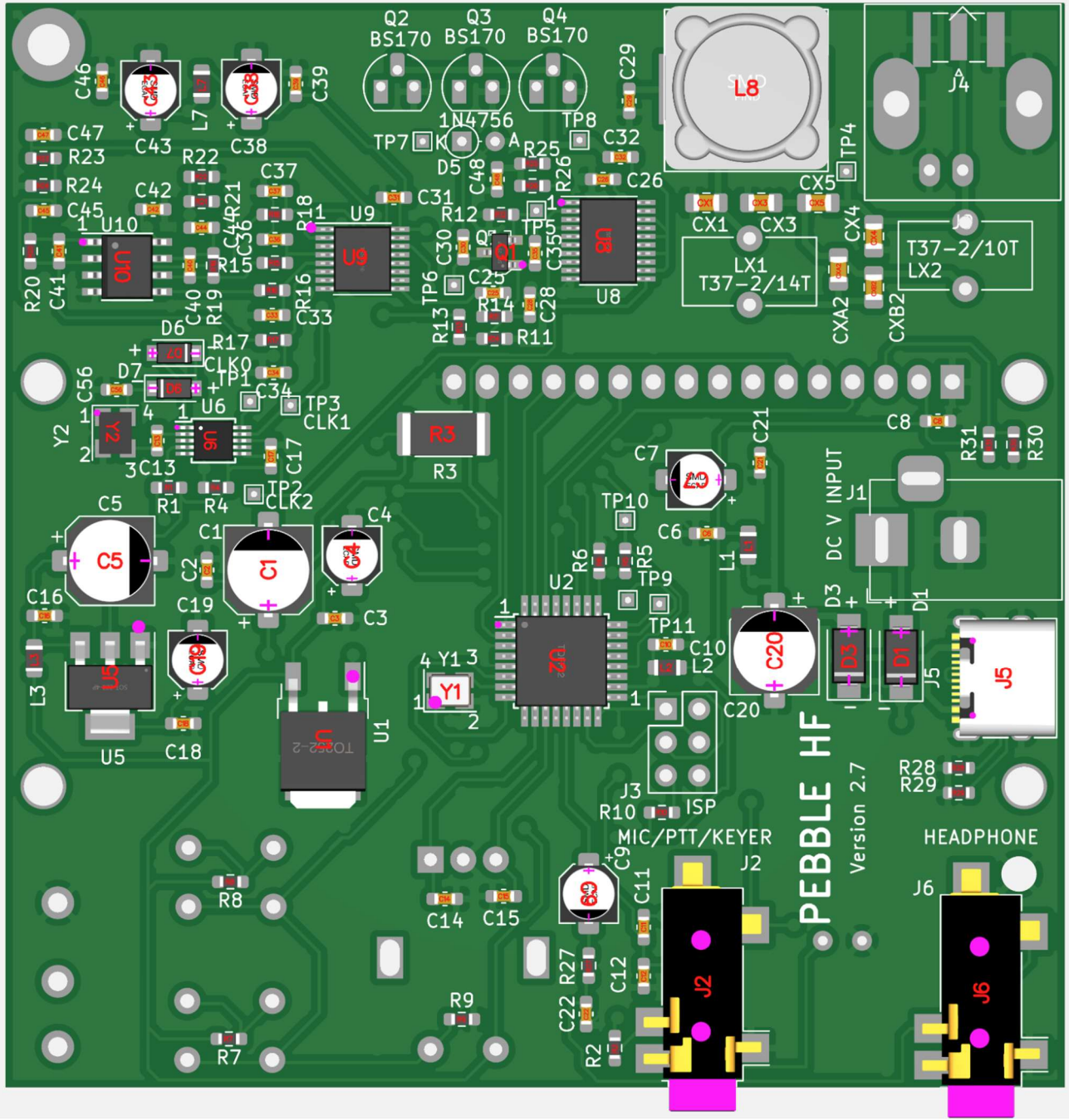
- ☐ Start by unpacking your kit, identifying each component, and checking the components against the BOM.

PCB and Schematic Diagram

Here is the layout of the component and rear side of the main 40m Pebble PCB ver 2.7.

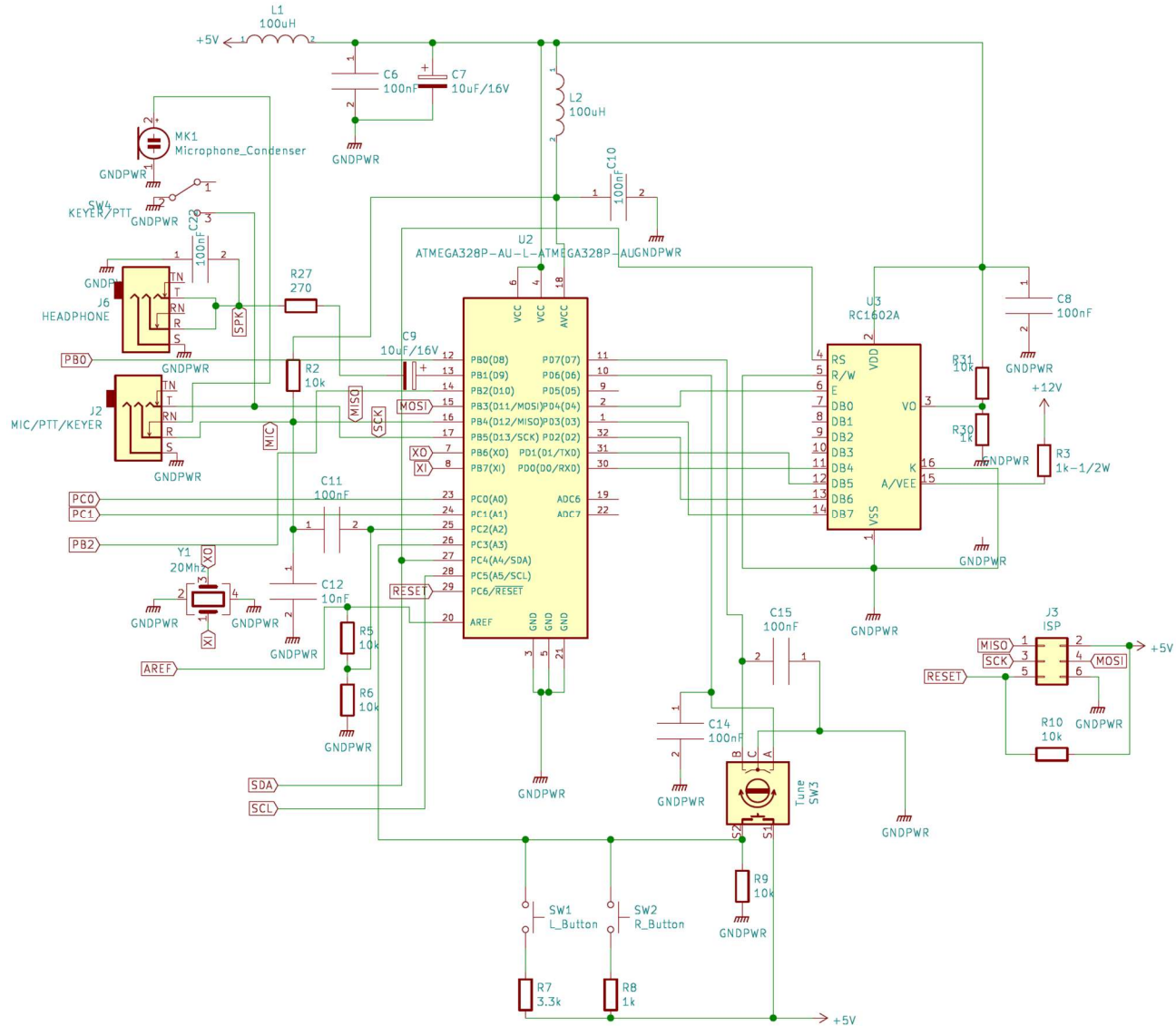


The part placement on the main PCB is shown below:

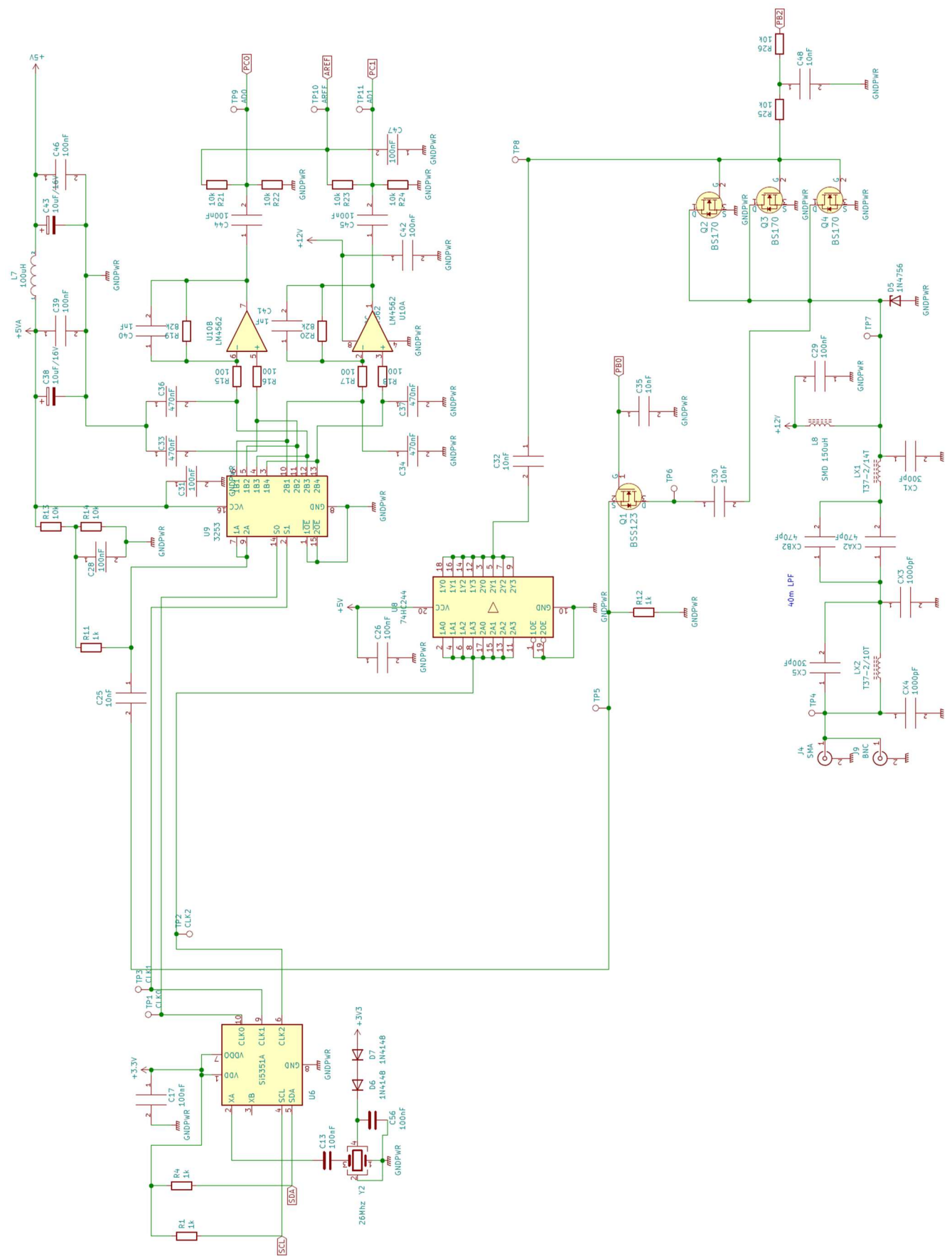


Schematic Diagram

The schematic diagram for the project design is shown below broken into two sections to provide better readability:



PEBBLE HF adapted from uSDX TRX Firmware and base schematic by PE1NNZ		
40m Single Band All Mode HF SDR Transceiver		
HAPPY-HACKY-WB2CBA		
Sheet: 7		
File: SDR_SSB.sch		
Title: PEBBLE HF		
Size: A2	Date: 12/5/2025	Rev: V2.7
KiCad E.D.A. eschema (5.1.12)-1	Id: 1/1	
10	11	



Building the 40m Pebble

Follow the 20m Pebble Assembly Instructions

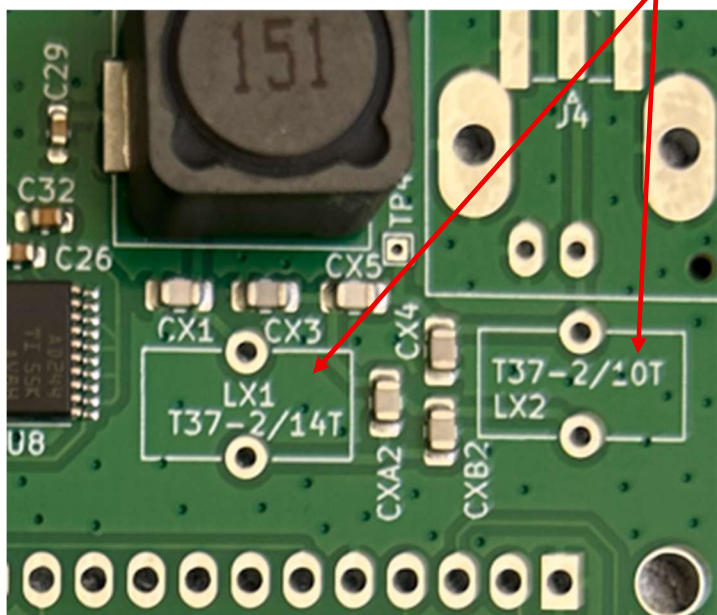
The building process for the 40m pebble is very similar to that for the standard 20m version, and you should start by downloading the 20m Pebble HF Assembly Instructions from here:

<https://pebblehf.com/docs.html>

Follow the assembly instructions for the 20m Pebble with the following exceptions:

40m LPF Inductor Winding

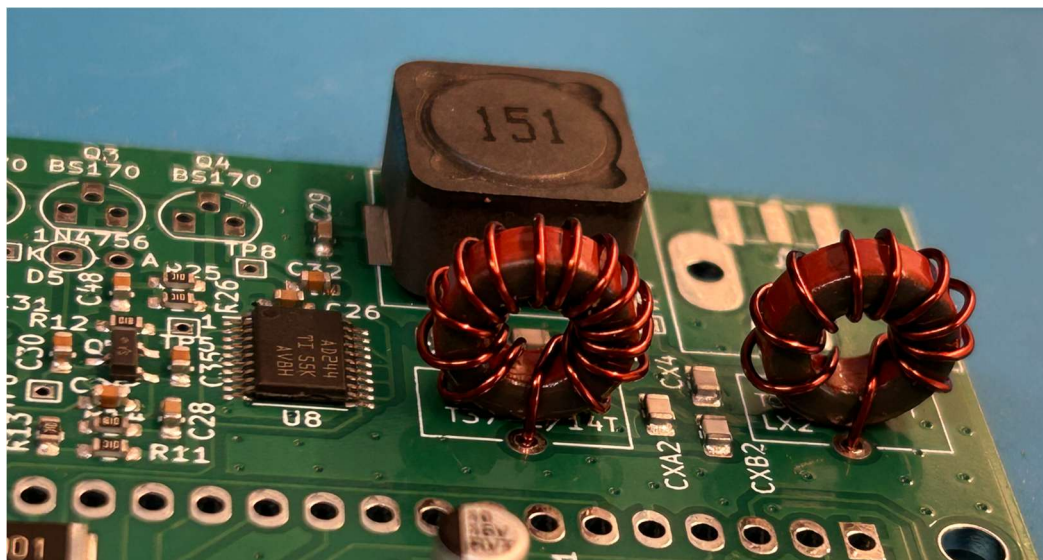
The 40m Pebble uses a different inductor arrangement for its low pass filters – whilst using the same T37-2's as the 20m version, the 40m variant uses a different number of windings on the toroids. The number of windings required are clearly identified on the 40m Pebble PCB as shown below:



LX1 requires turns (14T) on the T37-2 – this can be wound by measuring (twice) and then cutting a 230mm length of insulated wire from the 500mm length provided in the kit. Wind the inductor with 14 turns around the inductor using the same technique as that outlined in the 20m Pebble Assembly Instructions.

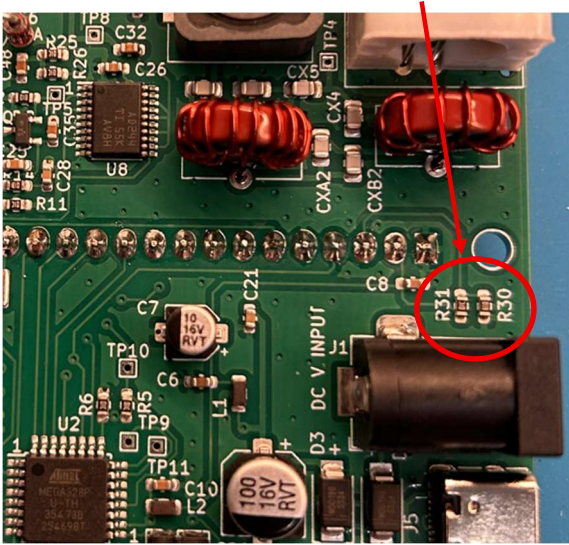
LX2 requires 10 turns (10T) on the T37-2. Use the remaining length of insulated wire to cut approx. 170mm of wire to wind the 10 turns onto the toroid using the same technique as outlined in the Assembly Instructions.

The finished windings will look like that shown below when installed.



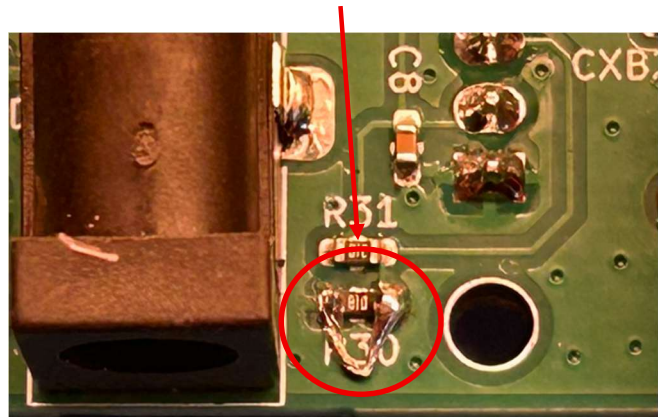
Contrast Adjustment - Resistor Modification R30

The version 2.6 Pebble PCB for 20m used a small variable resistor to adjust the voltage supplied to the LCD's #3 pin to adjust the contrast on the LCD. In the version 2.7 PCB, the variable resistor was removed and replaced by a voltage dividing network consisting of R30 (1K) and R31 (10K), with the resulting voltage supplied to the LCD which was suitable for the LCD displays being used in USA production runs.



However !!! the voltage required to adjust the contrast will vary between displays sourced from different manufacturers, and this is achieved by varying the R30 resistor. The main PCB is supplied with a 1K resistor for R30, and with the displays sourced for the MWRS build this resulted in insufficient contrast to make the screen readable.

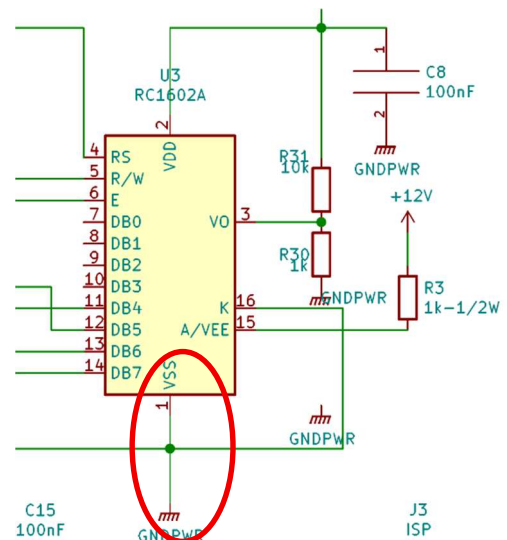
This situation was easily solved by shorting out R30 using a small off cut lead from one of the thorough hole components and soldering it to bridge R30 as shown below:



The resulting screen contrast will be similar to that shown on the left.

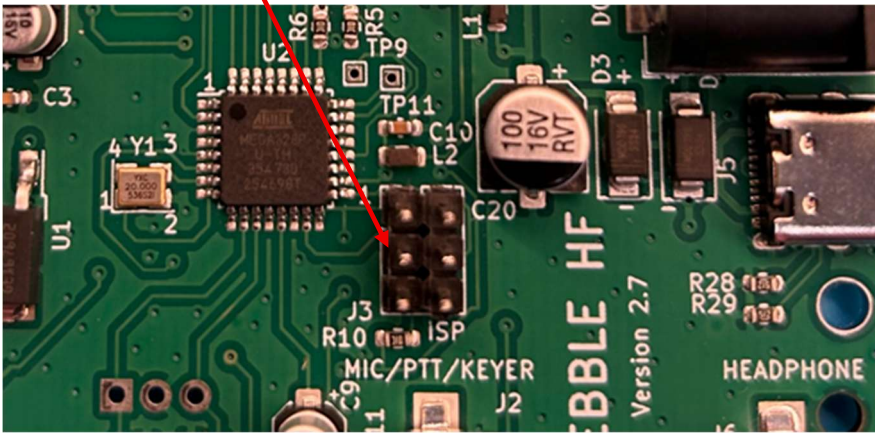


You may need to experiment with different values of R30 if future screen sources vary their #3 Pin contrast voltage requirements.



Installing the Program Header

The MWRS kit provides 2 x 3 pin 2.54mm spaced male header strips which need to be soldered in parallel to form the 6 pin programming header. Solder the two 3 pin strips in parallel as shown below ensuring that the pins are close together and vertical.



Installing the Pebble 40 Firmware

To operate the 40m Pebble you will need to install the bootloader and appropriate firmware. The firmware is available from my GitHub site and can be found here:

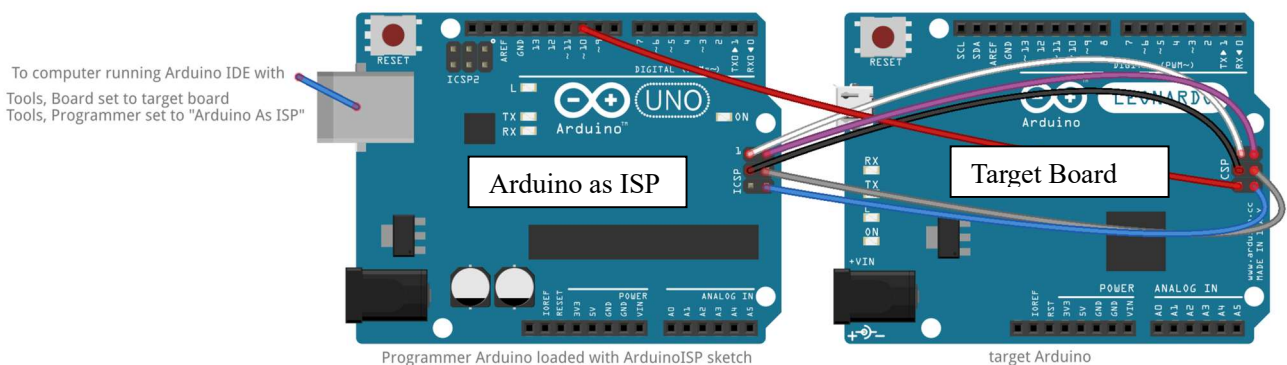
<https://github.com/VK2ARH/40m-Pebble-HF-Radio>

The bootloader and firmware are uploaded to the pebble using the Arduino IDE. There is a detailed guide covering the pebble firmware upgrade which can be found here:

<https://github.com/mspiceland/usdx-pebblehf/blob/main/PROGRAMMING.md>

Some may find the detail in the guide a little confusing, so I have detailed how I uploaded the firmware using the Arduino IDE using an Arduino as a ISP programmer, which I found to be the easiest method.

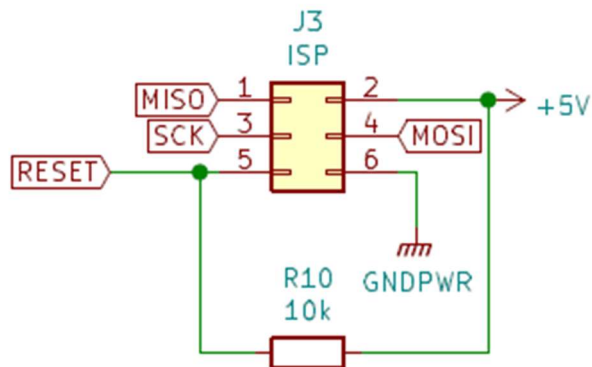
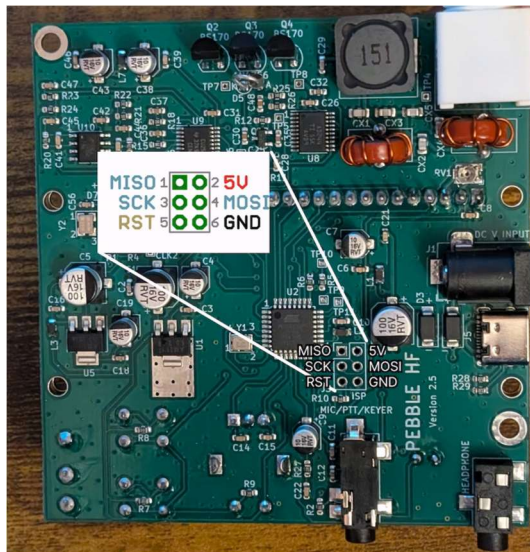
Programming the Pebble is a two stage process – we need to burn the bootloader to the board (only needs to be done once when first uploading firmware this step is not required if updating the firmware with a later release) and then upload the firmware to the Pebble. There are many internet sources covering the process of burning bootloaders to an Arduino, but this is the process I used:



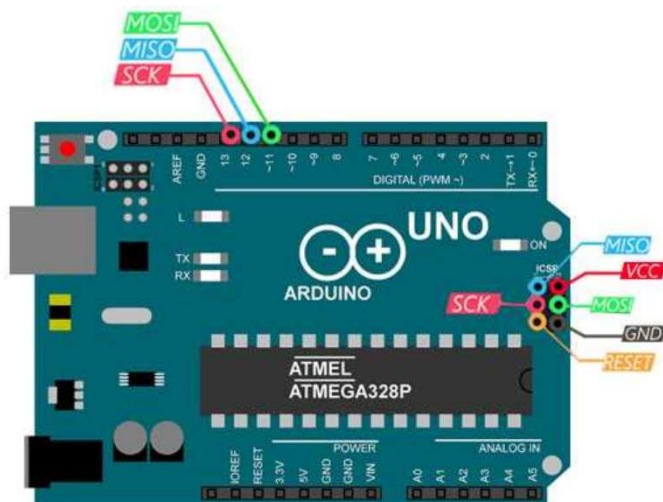
One Arduino (eg: Uno or Nano) is used to connect to the target board (Pebble) using the ISP port of the Arduino and the Programming Header on the Pebble. The above example shows one Uno being used as a programmer and the second target board as the board to be programmed. In our case we will be using this method with the target board being the Pebble, but the connections to the ISP plug remain the same.

Connections to the Pebble Program Header

To do this you connect the Arduino Uno to the Pebble using a six pin cable with the following connections:



The corresponding ISP connections on the Arduino Nano are shown below. You can either use individual ‘dupont’ connector / ribbon cable or a dedicated 6 pin to 6 pin cable, but ensure that when connected the pin connections match those shown below:



Arduino		Pebble
MISO	—	MISO
SCK	—	SCK
RESET	—	RESET
VCC	—	+5v
MOSI	—	MOSI
GND	—	GND

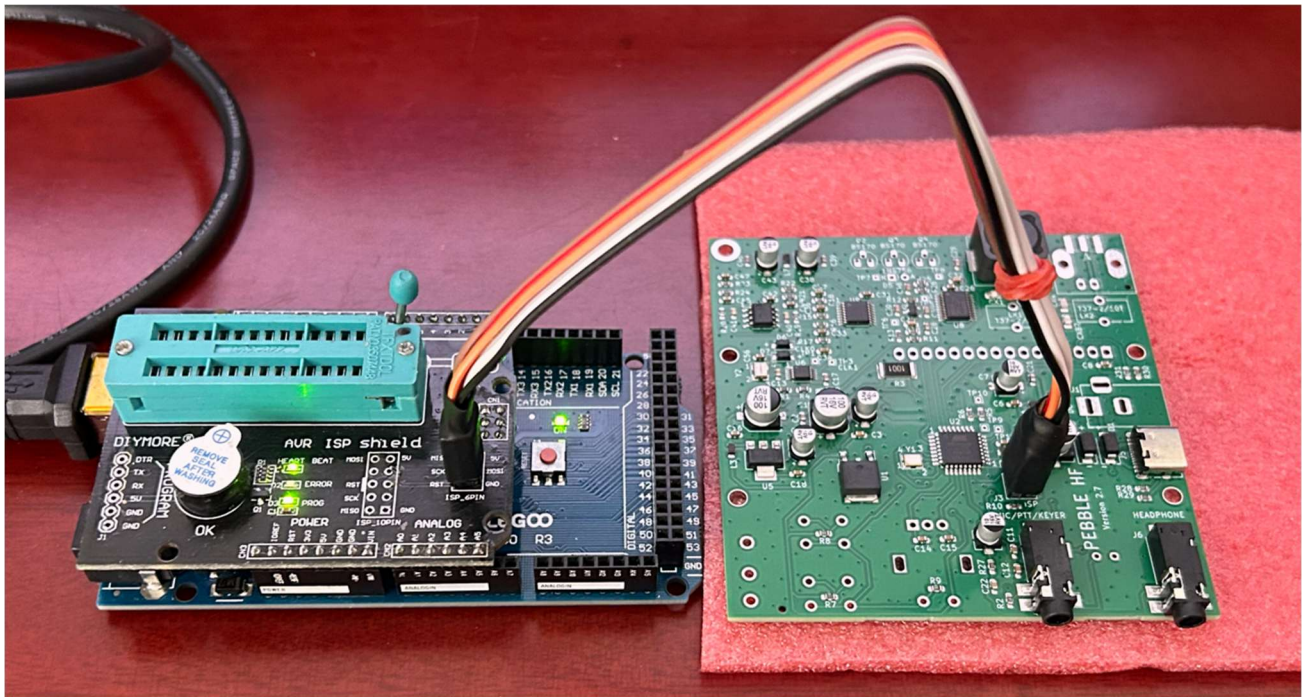


Dupont Female Connector Cables



6 pin Female ISP Connector Cable

This is how I programmed my own Pebble using 6 x dupont female connectors held together to form a plug with heat shrink. Although I am using ISP Shield on an Arduino Mega (as the ‘Arduino as an ISP’), the same ISP output plug is available on a native Arduino Nano or Arduino Uno boards.

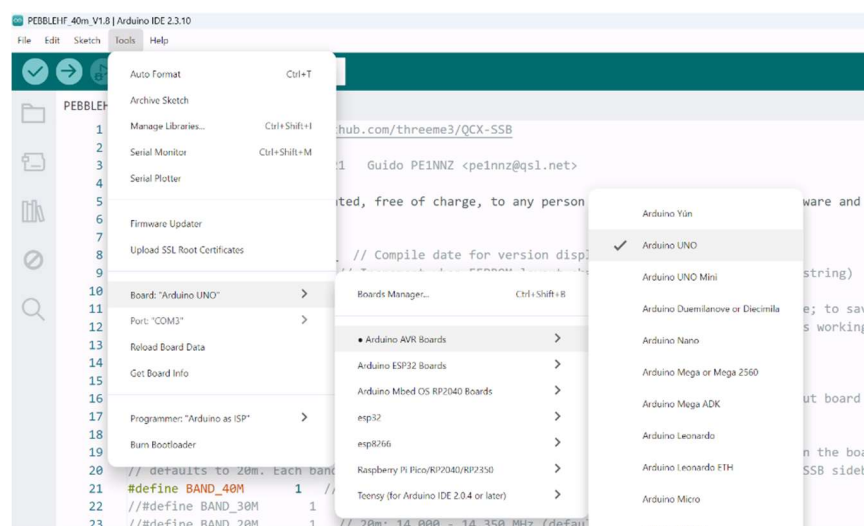


Programming the Pebble

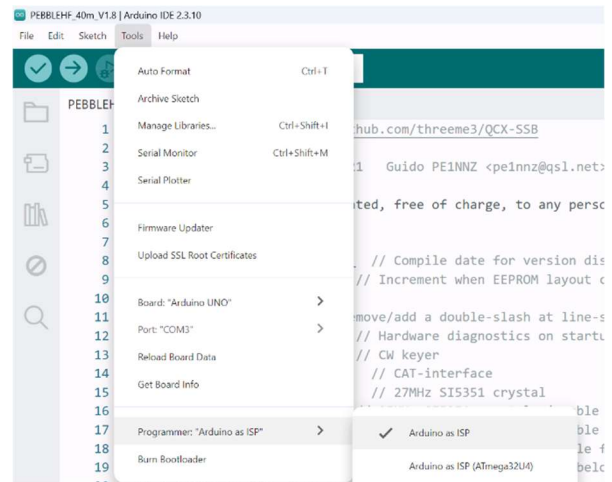
- When the Pebble is connected to the Arduino ISP pins (and double checked to ensure the correct pin connection has been made), connect the Arduino to your computer via USB.
- Open the Arduino IDE and load the Pebble Software.

Burning the Bootloader

- 1- Choose Arduino Uno as the board, and select the appropriate comm port for the Arduino under the tools drop down menu.

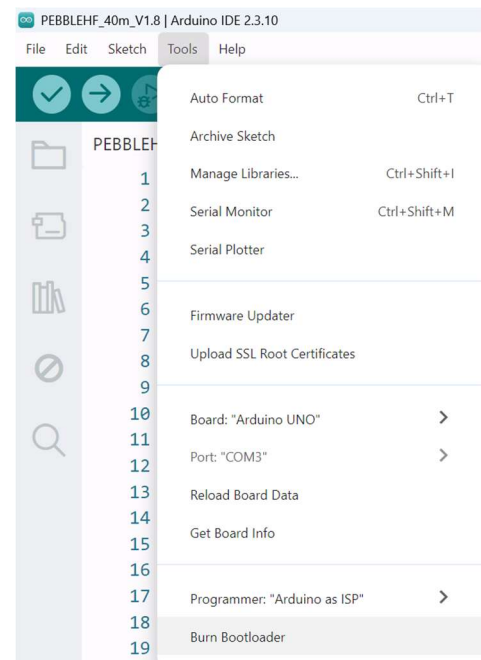


- 2- Choose the Programmer ‘Arduino as ISP’ under tools menu



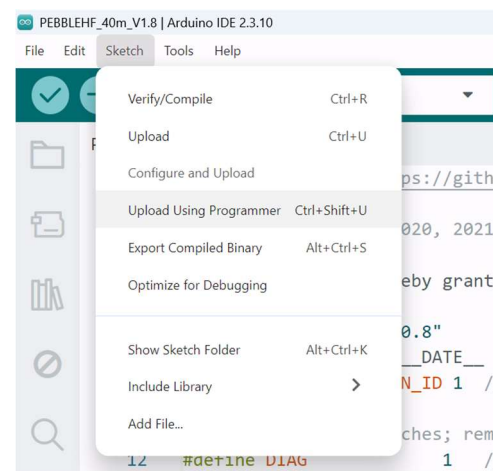
- 3- Burn the bootloader onto the chip on the board by clicking “Burn Bootloader” selected from the Tools drop down menu.

Note: if you are updating the firmware there is no need to burn the bootloader a second time and you jump to step 4 to upgrading the firmware without burning the bootloader again.



Uploading Firmware

- 4- Choose ‘Upload using programmer’ under sketch menu, and which will then upload the firmware to the Pebble.



Once the firmware has been completed and successfully uploaded, the Pebble will ‘reboot’ and you should be greeted with the Pebble main screen in the display, and be ready to commence testing and operation.

Placing your own Callsign on the Pebble Display

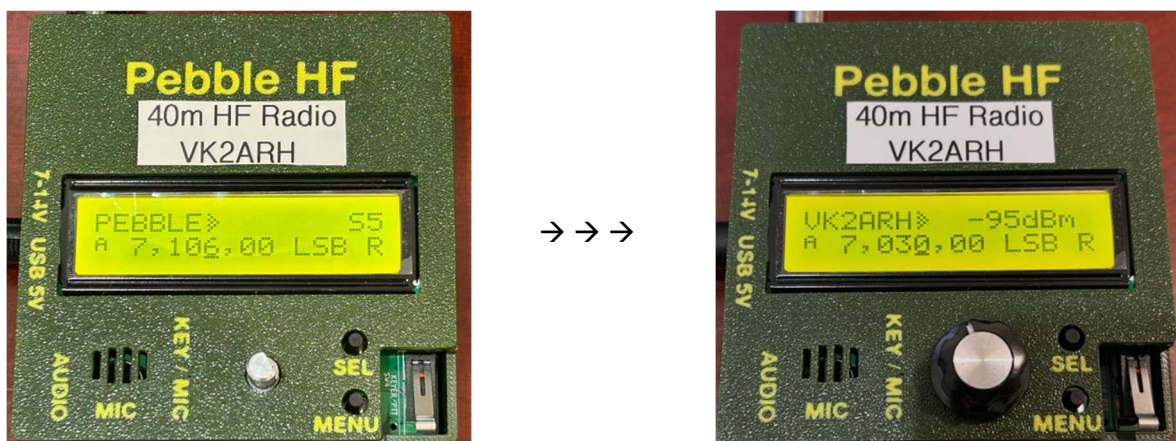
If you would like to display your own callsign rather than 'PEBBLE' on the main display, search for 'PEBBLE' in the firmware once loaded and replace 'PEBBLE' with your callsign as shown below. In the current version of the firmware this can be found at line 4214 of the code.

```
4213  #else
4214  | lcd.print(F("PEBBLE")); //Pebble HF single-band variant
4215  #endif //QCX
4216  | lcd.print('\x01'); lcd_blanks(); lcd_blanks();
4217  }
```

Replace "PEBBLE" with your callsign, and save the code before uploading

```
4213  #else
4214  | lcd.print(F("VK2XYZ")); //Pebble HF single-band variant
4215  #endif //QCX
4216  | lcd.print('\x01'); lcd_blanks(); lcd_blanks();
4217  }
```

The result is shown below – having changed the display to reflect my own callsign.



Before undertaking any power on testing, connect the Pebble to a 50 Ohm dummy load.

Familiarize yourself with the user interface and how to operate the Pebble. A full operating manual can be found here:

https://github.com/mspiceland/usdx-pebblehf/blob/main/OPERATING_MANUAL.md

I have also attached a field reference guide at the end of this manual.

Testing the Output Power

- ☐ Connect a QRP power meter (a RF Power meter capable of accurately reading RF power up to 10 watts) eg: Barb- a-Watt in series with a dummy load.
- ☐ Connect a 12v power source to the DC in socket. The socket uses a 5.5mm x 2.1mm DC plug center +ve.
- ☐ Change the mode to CW, and ensure that the CW Practice mode is turned off.
- ☐ Key down the onboard CW key – you should see over 4w of output power being transmitted into the dummy load. 5w will generally be observed with the input voltage around 13v. If powering with 5v supply using the USB-C port (do not use a PD USB-C supply) you will see around 1w of RF output – this is an ideal power source and power output for using the Pebble as a WSPR beacon.
- ☐ Use the ‘finger test’ by placing a finger on top of the BS170 transistors and monitor their temperature when key down. If the unit is performing correctly, you should observe only a slight warming of the BS170’s during a full 2 minute key down cycle with a 12v supply. If the transistors start to warm up (even to the point of being so hot that you cannot hold your finger on the transistors), check to see that you are transmitting into a 50 Ohm load without any shorts. High SWR or an incorrect match with the antenna will result in an increase in BS170 temperature – you should ensure that you always are transmitting into a tuned antenna system to prevent damage to the BS170’s and that the voltage supply remains within the required range.
 - *NOTE: If you do damage (burn out) the BS170 transistors they are easily and inexpensively replaced, but best not to damage them.*



Whilst this example shows 5w output with 13.1v input, you will typically see about 4.4w output when powered from a 12.6v LiPo battery which will draw around 560mA on Tx.

The photo above shows the harmonic suppression of the 40m Pebble demonstrating over 47db of suppression on the 2nd Harmonic

Toroid Winding

The following notes regarding Toroid windings are taken from the following resource,

[usdx-pebblehf/TROUBLESHOOTING.md](https://github.com/usdx-pebblehf/TROUBLESHOOTING.md) at main · mspiceland/usdx-pebblehf

Winding the toroids correctly is critical for getting the most out of your Pebble HF and for protecting your finals (RF transmitting transistors).

The Pebble HF uses a **Class E PA design**. This high-efficiency design is what allows the Pebble HF to operate without a heatsink. However, this efficiency depends on well-wound toroids.

Winding Tips

- **Pull each turn tight** — loose windings reduce efficiency
- **Space turns evenly** — consistent spacing improves performance
- **Inspect your work** — poor windings are the most common cause of issues

Expected Performance

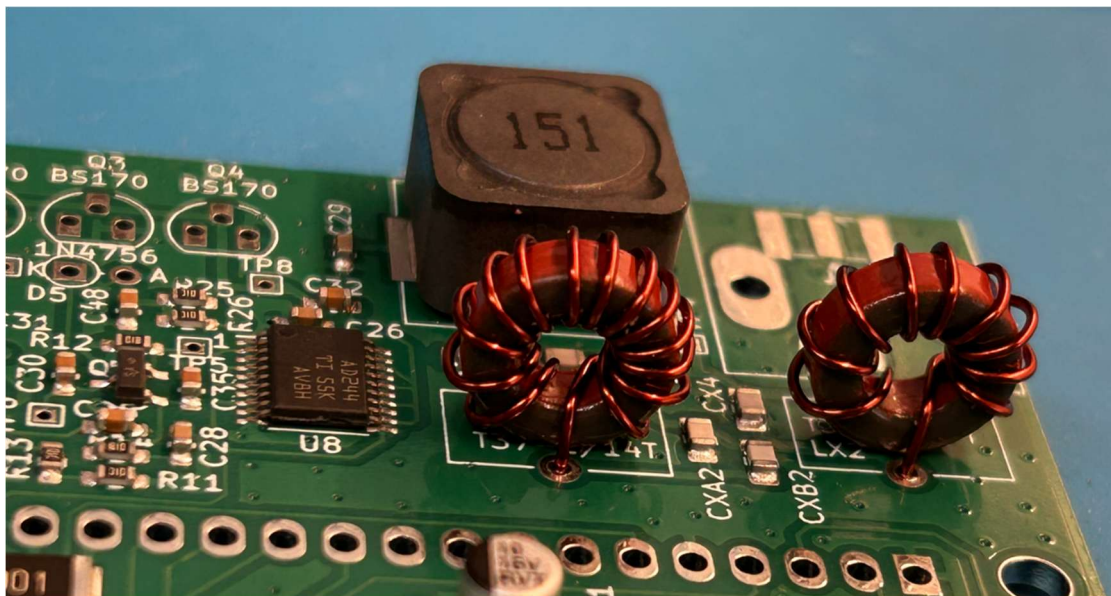
With proper toroid winding and 13.8V input:

Efficiency	Output Power	Notes
75–79%	4.5–5.5 W	Normal, healthy operation
Below 50%	Reduced	Excessive heat generated

Warning: If efficiency drops below 50%, the PA will generate considerable heat. This will eventually cause the BS170 transistors to fail after extended TX periods.

Protection

The Pebble HF includes diode protection against poor SWR and accidental transmission with no antenna connected. **The most likely cause of failed transistors is improper toroid winding**, not SWR issues.



3D Printed Housing

There is a lot of scope for the development of 3D printed cases to house the Pebble, and our 3D printing SIG (Special Interest Group) are active with the development of various housing options either as the basic case for the main board or future housings which may include amplified speakers, batteries etc.

In the meantime Ham Radio Duo's website is a great resource and includes all the files and instructions to enable you to print your own 3D printed housing for the Pebble. Details can be found here"

<https://hamradioduo.com/print-your-own-pebble-hf-case/>



Simon VK2YU has also developed the files to enable a specific 40m Version of the enclosure which incorporates your own callsign printed on the case. The files required and instructions to print the VK2YU variant can be found here:

<Insert Link to Simon's Website>

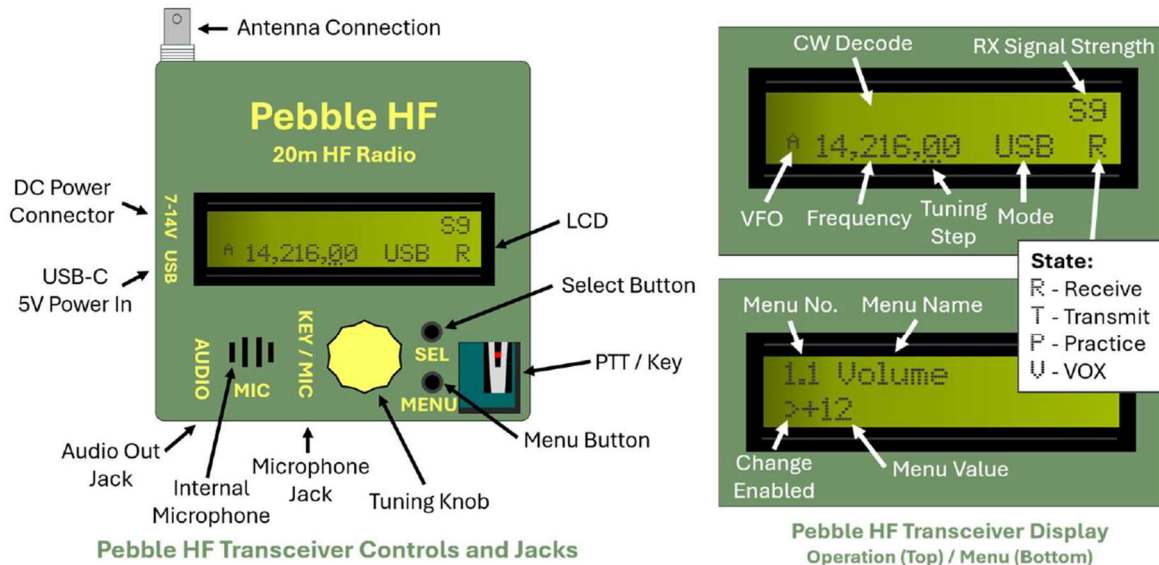
Here is a first look at Simon's enclosure variant with a couple of colour options:



Pebble HF Field Reference Guide

By Steve KV4AN – original can be downloaded along with other documentation and resources from:

<https://pebblehf.com/docs.html>



Pebble HF Operating Instructions

Power On/Off

- Connect power to turn On
- Press and Hold *Tuning Knob* and rotate to “-1” to turn transceiver Off

Volume

- Press and Hold *Tuning Knob* and rotate for volume level “-1” (power Off) to 16

Frequency

- Rotate *Tuning Knob* to change frequency
- Short Press *Tuning Knob* to change Tuning Step

Mode

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.2 Mode
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to mode (use “CW” or “USB”)
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

VFO A/B

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.6 VFO Mode
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to “A” or “B”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

RIT On/Off

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.7 RIT
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to “On” or “Off”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Filter Bandwidth

- Double Click *Select Button* to cycle through bandwidth settings: “FULL”, “3000”, “2400”, “1800” (SSB) or “500”, “200”, “100”, “50” (CW)

Pebble HF Operating Instructions

AGC On/Off

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.8 AGC
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to “ON” (SSB) or “OFF” (CW)
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Noise Reduction

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.9 NR
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to adjust level “0” (OFF) to “8”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Attenuator (Analog, Front-End)

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.10 ATT
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to level “0” (OFF) to “-73” dB
Typical values used are “0” or “-13”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Attenuator (Digital, Filter Stage)

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 1.11 ATT2
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to level from “0” (OFF) to “16”
Normal setting is “0” or “1”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

CW Decoder On/Off

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 2.1 CW Decoder
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to “ON” or “OFF”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Sidetone Volume

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 2.3 CW Side Vol
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* from “-1” (mute) to “16”
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Keyer Speed

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 2.5 Keyer Speed
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* from “1” to “60” WPM
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

Keyer Mode

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 2.6 Keyer Mode
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* to “Iambic A”, “Iambic B”, or “Straight”

Use “Straight” when external key is not plugged in.

- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

TX Drive (Transmit Audio / Power Level)

- Short Press *Menu Button*, rotate *Tuning Knob* to Menu 3.3 TX Drive
- Short Press *Menu Button* to enable change, rotate *Tuning Knob* from “0” to “8”
Typical settings are “4” for SSB and “8” for CW/Digital
- Short Press *Select Button* to save change (short press *Select Button* again to exit menu)

See the Operator Manual for more Information and Menu Options

Setup for Digital Modes

Details reproduced here from the following GitHub Site:

[usdx-pebblehf/TROUBLESHOOTING.md at main · mspiceland/usdx-pebblehf](https://github.com/mspiceland/usdx-pebblehf/blob/main/TROUBLESHOOTING.md)

For digital modes (FT8, JS8Call, etc.), you will need an external USB sound card.

Connections

Sound Card Jack	Connects To
Input (red)	Pebble HF headphone output
Output (green)	Pebble HF mic/key input

Recommended Settings

1. **Firmware:** Use v1.0.4 or later (fixes sound card compatibility issues)
2. **VOX:** Set to **ON**
3. **Noise Gate:** Set to **50** (increase if you experience unexpected TX)
4. **Practice Mode:** Start with **ON** to verify timing before transmitting

Setup Procedure

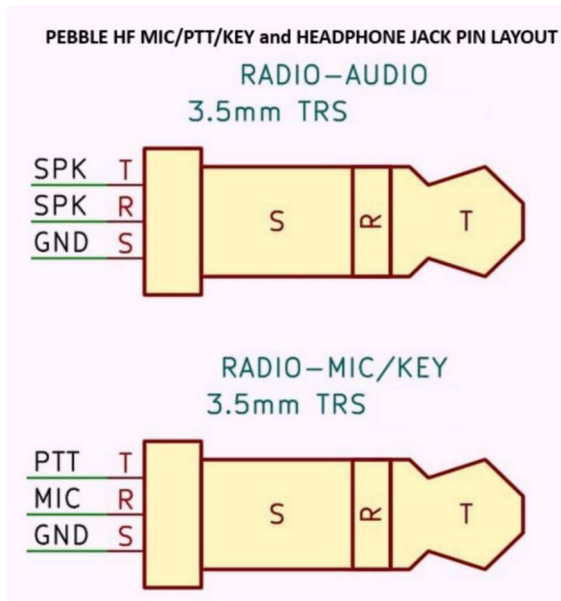
1. Connect the sound card as shown above
2. Set **Practice Mode** to **ON**
3. Configure your digital mode software (WSJT-X, JS8Call, etc.) to use VOX for the radio
4. Verify the radio keys at the correct times
5. Set **Practice Mode** to **OFF** to transmit for real

Sound Card Compatibility Note

Starting with firmware v1.0.4, when VOX mode is enabled, the Tip conductor from the TRS (tip-ring-sleeve) audio cable is ignored. Some sound cards have impedance characteristics that would cause the radio to transmit as if the PTT button was pressed. This change improves compatibility with a wider range of external sound cards.

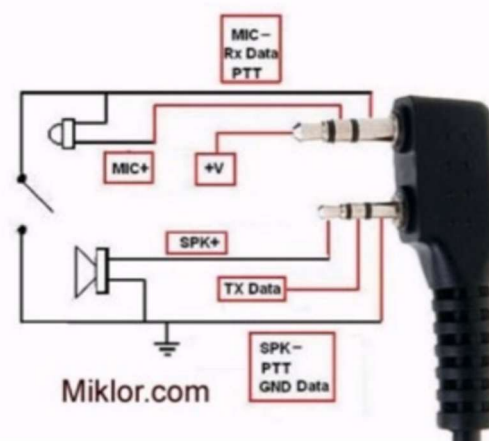
Microphone and Speaker Plug Connections

The pebble's audio output socket is capable of driving an earpiece but not a larger speaker. Use of an external amplified speaker enhances the user experience of the Pebble and is easily connected using a 3.5mm stereo plug to the Audio output socket.



The onboard electret microphone works surprisingly well – however should you wish to connect an external microphone to the Pebble this can also be achieved by the use of a 3.5mm stereo plug. The Connection diagram for external speaker and microphone is shown on the left.

If you wish to connect a standard K1 plug used by Baofeng or Kenwood radio the pin configuration or this connection is shown below:



I found the following diagram useful when wiring up a jumper cable for my Retekis HK008 speaker microphone for use with the Pebble.

The K1 plug 2.5mm plug (audio) is converted to a 3.5mm stereo plug with no change in the wiring.

The K1 plug 3.5mm Microphone connection requires the Microphone and PTT lines to be switched, and the ground connected to the audio ground line as shown on the following diagram.

